

Title of the Project

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology

in

Electrical and Electronics Engineering

by

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23BEE0XXX

Under the guidance of

Prof. / Dr.

School of Electrical Engineering

VIT,Vellore.



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Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

May 2026

DECLARATION

I here by declare that the thesis entitled “**Thesis Title**” submitted by me, for the award of the degree of *Bachelor of Technology in Electrical and Electronics Engineering* to VIT University is a record of bonafide work carried out by me under the supervision of **Dr. Guide Name**, Designation, School or Centre Name, VIT University, Vellore.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Vellore

Date:

Signature of the Candidate

CERTIFICATE

This is to certify that the thesis entitled “**Thesis Title**” submitted by **Student Name(Reg. No)**, School or Centre, VIT, Vellore, for the award of the degree of *Bachelor of Technology in Electrical and Electronics Engineering* , is a record of bonafide work carried out by him/her under my supervision during the period, xx.xx.2023 to xx.xx.2024,, as per the VIT code of academic and research ethics.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university. The thesis fulfills the requirements and regulations of the University and in my opinion meets the necessary standards for submission.

Place: Vellore

Date: dd/mm/yyyy

Signature of the Guide
(Dr. Guide Name)

The thesis is satisfactory / unsatisfactory

Approved by

Head of the Department
EEE/EIE

Dean,SELECT

Acknowledgements

With immense pleasure and deep sense of gratitude, I wish to express my sincere thanks to my supervisor **Dr. Guide Name**, Designation, School or Centre Name, VIT University, without his/her motivation and continuous encouragement, this project work would not have been successfully completed.

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I express my sincere thanks to **Dr. Dean / Director Name**, Dean/Director, School or Centre, VIT University for his kind words of support and encouragement. I like to acknowledge the support rendered by **my colleagues** in several ways throughout my project work.

I wish to extend my profound sense of gratitude to **my parents** for all the sacrifices they made during my research and also providing me with moral support and encouragement whenever required.

Last but not the least, I would like to thank the lab in charge **Name** for their constant encouragement and moral support along with patience and understanding.

Place: Vellore

Date: dd/mm/yyyy

Student Name

Executive Summary

Executive Summary should be one page summary the thesis typed one and a half line spacing, Font Style: Times New Roman and Font Size: 12. The executive summary is a very brief summary of the thesis contents. It should be about one page long not more than 200 words. The 200-word statement should describe the problem addressed by your thesis, a description of the work completed and a summary of any findings or lessons learned.

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LIST OF TERMS AND ABBREVIATIONS

F_a Force	1
CSK Chennai Super King	1
LQR Linear Quadratic Regulator	1
MPC Model Predictive Control	1
URL Uniform Resource Locator	3, 6

CHAPTER 1

INTRODUCTION

1.1 Objective

A Capstone Project Report/thesis or dissertation is a document submitted in support of candidature for an academic degree or professional qualificationChennai Super King (CSK) presenting the author's research and findings. Force (F_a) In some contexts, the word "thesis" or a cognate is used for part of a bachelor's or master's course, while "dissertation" is normally applied to a doctorate, while in other contexts, the reverse is true.

These guidelines are provided to formally expose you to the various ethical and technical issues involved in writing up your work and the format you are required to adhere to while submitting your work as B.Tech/M.Tech /Capstone Project Report/ Thesis.

Project objective is the goal that your project is supposed to achieve at the end of your project. It should be specific, measurable, achievable, A linear quadratic regulator relevant and time-bound.It should also readable and understandable.

1.2 Motivation

(Das et al. 2018)Knowing the difference between ethical and unethical practices in technical writing requires an understanding of plagiarism,Linear Quadratic Regulator (LQR) paraphrasing, and quotation. These concepts are defined below,Model Predictive Control (MPC). The definitions are reproduced from the 'Handbook of Technical Writing' by Brusaw.

The motivational statement about the project should be what has done previously related to this project,that to make you want to do the project that you are interested in, what you hope to get out of the project , and what you want to apply new thing in the project.

1.3 Background

To use someone else's exact words without quotation marks and appropriate credit, or to use the unique ideas of someone else without acknowledgement, is known as plagiarism. In publishing, plagiarism is illegal; in other circumstances, it is, at the least, unethical. You may quote or paraphrase the words or ideas of another if you document your source.(Mendiratta et al. 2017) Although you need not enclose the paraphrased material in quotation marks, you must document the source.

Paraphrased ideas are taken from someone else whether or not the words are identical. Paraphrasing a passage without citing the source is permissible only when the information paraphrased is common knowledge in a field. (Common knowledge refers to historical, scientific, geographical, technical, and other type of information on a topic readily available in handbooks, manuals, atlases and other references).

A background of a project is just a simple and short statement of the project, meaning why we need to initiate it and what problems and needs will be addressed once it's been implemented successfully.

CHAPTER 2

PROJECT DESCRIPTION AND GOALS

2.1 Review of Literature

A literature review is a text of a scholarly paper, which includes the current knowledge including substantive findings, as well as theoretical and methodological contributions to a particular topic. Literature reviews are secondary sources, and do not report new or original experimental work. Most often associated with academic-oriented literature, such reviews are found in academic journals, and are not to be confused with book reviews that may also appear in the same publication. Literature reviews are a basis for research in nearly every academic field (Maedche and Staab 2001). A narrow-scope literature review may be included as part of a peer-reviewed journal article presenting new research, serving to situate the current study within the body of the relevant literature and to provide context for the reader. In such a case, the review usually precedes the methodology and results sections of the work.

Producing a literature review may also be part of graduate and post-graduate student work, including in the preparation of a thesis, dissertation, or a journal article. Literature reviews are also common in a research proposal or prospectus (the document that is approved before a student formally begins a dissertation or thesis) (Doan et al. 2002, Duzdevich et al. 2014).

2.2 Project Description

The main types of literature reviews are: evaluative, exploratory, and instrumental (Duzdevich et al. 2014, Thiruganam et al. 2010).

A fourth type, Uniform Resource Locator (URL)the systematic review, is often classified separately, but is essentially a literature review focused on a research question, trying to identify, appraise, select and synthesize all high-quality research evidence and arguments relevant to that question. A meta-analysis is typically a systematic review using statistical methods to effectively combine the data used on all selected studies to produce a more reliable result.

2.2.1 Software

Gilbarg and Trudinger (2015) distinguish between the process of reviewing the literature and a finished work or product known as a literature review. The process of reviewing the literature is often ongoing and informs many aspects of the empirical research project. All of the latest literature should inform a research project. Scholars need to be scanning the literature long after a formal literature review product appears to be completed.

2.2.2 Hardware

Gilbarg and Trudinger (2015) distinguish between the process of reviewing the literature and a finished work or product known as a literature review. The process of reviewing the literature is often ongoing and informs many aspects of the empirical research project. All of the latest literature should inform a research project. Scholars need to be scanning the literature long after a formal literature review product appears to be completed.

2.3 Goals

(Das et al. 2018) A careful literature review is usually 3 to 5 pages and could be longer. The process of reviewing the literature requires different kinds of activities and ways of thinking (Duzdevich et al. 2014) and Kothari (2004) link the activities of doing a literature review with Benjamin Bloom's revised taxonomy of the cognitive domain (ways of thinking: remembering, understanding, applying, analysing, evaluating, and creating).

A Project Description is a document that outlines the details of a specific project in a structured format covering all stages of the project and the processes involved in it and goal refers to the desired outcome of the project.

CHAPTER 3

TECHNICAL SPECIFICATION

3.1 Software Specification

”When you paraphrase a written passage, you rewrite it to state the essential ideas in your own words. Because you do not quote your source word for word when paraphrasing, it is unnecessary to enclose the paraphrased material in quotation marks. However, the paraphrased material must be properly referenced because the ideas are taken from someone else whether or not the words are identical.

(Aggarwal 2013)Ordinarily, the majority of the notes you take during the research phase of writing your report will paraphrase the original material. Paraphrase only the essential ideas. Strive to put original ideas into your own words without distorting them.”

A technical specification is a detailed and comprehensive document that describes all technical procedures related to the project. It covers all the software and hardware information about the process of project completion.

3.2 Hardware Specification

”When you have borrowed words, facts, or idea of any kind from someone else’s work, acknowledge your debt by giving your source credit in footnote (or in running text as cited reference). Otherwise, you will be guilty of plagiarism. Also, be sure you have represented the original material honestly and accurately. Direct word to word quotations are enclosed in quotation marks.”

When you use programs written by others with or without modifications, the report/thesis must clearly bring this out with proper references, and must also reflect the extent of modification introduced by you, if any. A modified program is not entirely yours. Only a program, which you write from scratch, does not require source to be identified. Identification of source in all other cases is must. Standard subroutines (even if public domain) used in your programs must be properly referenced. Although programs need not be appended to the thesis, they must be submitted to your research supervisor in hard copy and other media. Inclusion of a computational flow chart in

your thesis is highly recommended, however.

The material presented in the thesis/report must be self contained. A reader must be able to reproduce your experimental, theoretical, computational, and simulations results based on the information presented in the thesis. You must mention the names of the suppliers whose chemicals/instruments were used in the work to allow a reader to setup an experiment. While discussing issues related to computation time, the hardware used must be specified accurately, using processor speed, etc.

3.2.1 Quotation and Reference to Earlier Work

If reproduction of some text material available in a published work can enhance the value to your thesis, you can add it to your thesis in the form of quoted material or a quotation. Such material should be indented on both sides over and above the indentation used for the regular text. It should preferably be single spaced, and appear as a separate paragraph(s), whether short or long. The idea is to make such material stand out from the rest of the text that you have written. Clearly, too many quotations or quoted paragraphs are not desirable in a thesis which is an original piece of work. Not quoting a material taken verbatim from another source is however plagiarism. Paraphrasing and giving credit to the author(s) is more accepted way of referring to earlier works.

3.2.2 Use of Abbreviations

In Chapters, while introducing abbreviations, follow:

```
\newacronym{URL}{URL}{Uniform Resource Locator}
```

and in chapters it can be used as:

```
\gls{URL}
```

Which produces the following output first time when you call it: URL and simply URL each subsequent time.

3.3 References

Choose a respected journal in your field in which title of the paper also appear in the list of references and consistently follow the citation style used by this journal. Names of all URL the authors with their initials, title of the article, names of editors for edited books or proceedings, and the range of pages that contain the referenced material must

appear in the bibliography. You should not mix citation styles of several journals and not to create your own style.

3.3.1 Citation Format

All references and citation should be of the standard "Harvard Style" (Author, Year)' format.

3.3.1.1 Single Author Citation

- Citation at the beginning of the sentence
 - Jones (2011) emphasized that citations in a text should be consistent.
- Citation at the End of the Sentence:
 - It was emphasized that citations in a text should be consistent (Jones, 2011).

3.3.1.2 Double Authors Citation

- Citation at the beginning of the sentence
 - Jones and Baker (2011) emphasized that citations in a text should be consistent.
- Citation at the End of the Sentence
 - It was emphasized that citations in a text should be consistent (Jones and Baker, 2011).

3.3.1.3 More than Two Authors Citation

- Citation at the beginning of the sentence
 - Jones et al. (2011) emphasized that citations in a text should be consistent.
- Citation at the End of the Sentence
 - It was emphasised that citations in a text should be consistent (Jones et al., 2011).

3.3.1.4 Sources Written in the same year by the same author(s)

- It was emphasized that citations in a text should be consistent (Jones, 1998a). In a work published later that year, Jones (1998b) proposed that...
- It was emphasized that citations in a text should be consistent (Jones, 1998a; 1998b).

3.3.1.5 Sources Written by same author(s) in different Year(s)

- (Smith, 2013; 2005; 2001)

CHAPTER 4

DESIGN APPROACH AND DETAILS (as applicable)

4.1 Design Approach / Materials and Methods

In this section describe about propose model of the project/ mathematical modelling.

4.2 Codes and Standards

In this section describe the standards which are considered in this project.

:Example IEEE 519 Standards for total harmonic voltage distortion.

4.3 Constraints, Alternatives and Tradeoffs

In this section describe about the design constraints and limitation, the tradeoff between the software and hardware implementation.

4.3.1 Constraints

4.3.2 Alternatives

4.3.3 Tradeoffs

The sequence in which the thesis should be arranged and bound should be as follows:

1. Cover Page and Title Page
2. Declaration
3. Certificate
4. Acknowledgement (1 page)
5. Executive Summary (not exceeding 200 words)
6. Table of Contents
7. List of Figures / Exhibits / Charts
8. List of Tables

9. List of Abbreviation
10. Chapters
11. Appendices
12. References

4.4 Page Dimensions and Binding Specifications

The dimension for printing the thesis in bond sheet (Size A4) is as follows:

4.4.1 Page Specification

Left Margin : 1.5 inch
Right Margin : 1 inch
Top Margin : 1 inch
Bottom Margin : 1 inch

4.4.2 Thesis Binding

The thesis should be bound using flexible cover (soft binding) for initial submission. The cover should be printed in black letters and the text for printing should be identical. After defensive viva, the thesis should be bound using rough surface rexin black cover printed in golden letters. The guidelines total number of pages in the report for each program is shown below:

Table 4.1 Total Number of Pages in the Thesis

Course	Color of Art Paper	Total Number of Pages in the Thesis
M.Tech	White	Min. of 65 (Excluding References, Appendices and Front pages)
B.Tech	White	Min. of 45 - Max. of 75 (Excluding References, Appendices and Front pages)

4.4.3 Printing Standard

For Initial Submission : JK Bond Sheet
For Final Submission : Executive Bond Sheet

4.5 Page Numbers

All text pages as well as program source code listings should be numbered using Arabic numerals at the bottom center of the pages.

4.6 Font

Times New Roman 12pt font should be used consistently throughout the text. Captions for tables and figures can be in smaller fonts, but not smaller than 10pt.

4.7 Paragraphs

No paragraph should have its opening line at the bottom of a page. A clear, consistent, but not too large a separation must be provided between the paragraphs throughout the thesis.

4.8 Line Spacing

The line spacing used should be the same throughout the text, and can be chosen to be 1.5. The lines in captions for figures and tables, Table of Contents, List of Figures, and List of Tables should be 1.5 line spacing.

4.9 Headings

Chapter Heading	Font Size: 16, Bold, Times New Roman, Title Case
Section Heading	Font Size: 14, Times New Roman, Title Case
Subsection Heading	Font Size: 12, Bold, Times New Roman, Title Case

CHAPTER 5

SCHEDULE, TASKS AND MILESTONES

5.1 Schedule

In this section give the complete description about the schedule of the project work.

5.2 Tasks and Milestones

In this section give the complete description about the task of the project work review wise and milestone achieved.

- All Table Caption should be in Title Case, TNR 10 Pt. It should be of the Format:
 - Table 1.1 Results of the Experiment ... (Centered)
- It should be cited as Table 1.1.
- Caption should appear above the Table.
- Table Header and the entries should be of Font TNR 10 Pt, Justified.
- For wider Table, the page orientation can be Landscape.
- For Larger Table, it can run to pages and the header should be repeated for each page of the Table.
- Table must be adjusted to fit in the page and no single row is left out for a new page.

Sample Table 5.1 and Table 5.2 are given below for your reference,

5.3 Math Equation in Thesis

All equation should be written using equation editor or using an equivalent tool.

- Equations should be numbered as : 1.1, 1.2 ...

Table 5.1 Country List

Country Name or Area Name	ISO ALPHA 2 Code	ISO ALPHA 3 Code	ISO numeric Code
Afghanistan	AF	AFG	004
Aland Islands	AX	ALA	248
Albania	AL	ALB	008
Algeria	DZ	DZA	012
American Samoa	AS	ASM	016
Andorra	AD	AND	020
Angola	AO	AGO	024

Table 5.2 Data Units, Sources, and Dates

Variable	Dates	Units	Source
Nominal Physical Capital Stock	1950-1990	Billions US\$	Nehru and Dhareshwar (1993)
Total Population	1950-1990	Billions	Nehru and Dhareshwar (1993)
Nominal GDP	1950-1990	Billions US\$	PWT
Real GDP per capita	1950-1990	2005 US\$ per capita	PWT

- Equation should be Centered, 12 Pt, TNR.
- Equation number should be right Justified
- It should be cited as Eqn. 1.1.

For example in Eqn. 5.1, The well known Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents. Meaning the next equation has no integer solutions:

$$x^n + y^n = z^n \quad (5.1)$$

The mass-energy equivalence is described by the famous equation in Eqn. 5.2

$$E = mc^2 \quad (5.2)$$

discovered in 1905 by Albert Einstein.

CHAPTER 6

PROJECT DEMONSTRATION

6.1 Figure

- All Figure Caption should be in the Title Case, TNR 10 pt, and it should be of the Format: Fig Chapter Number.Figure Number Figure Caption
- It should be cited as Fig. 6.1 Caption must appear below the Figure.
- For Smaller: (4 Figures arranged in Two Columns) / Page; Portrait Mod.
- For Medium: (2 Figures arranged one below the other / Page; Portrait Mode.
- For Larger: 1 Figure / Page; Landscape Mode.
- Figure Label should be in Font TNR 10 pt, Bold.
- Figure Resolution should be minimum of 300 DPI.

For example, The universe is immense and it seems to be homogeneous, in a large scale, everywhere we look at.



Fig. 6.1 Sample Picture of Universe

There's a picture of a galaxy above in Fig. 6.1

6.2 Graph

Another example graph in image format is given below Fig. 6.2

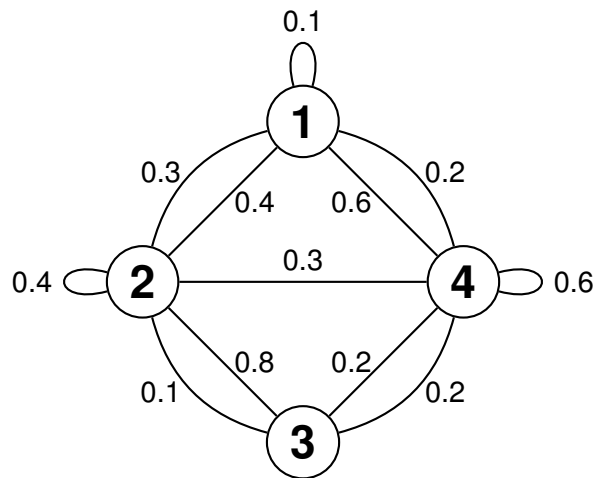


Fig. 6.2 Sample Graph

In this section provide the complete simulation and hardware results with description

CHAPTER 7

COST ANALYSIS / RESULT AND DISCUSSION (as applicable)

7.1 Cost Analysis

This section mention the cost involve to carried out the project(Hardware) or for software

Example for software project: Since the entire project was executed on MATLAB/Simulink, this component is free of cost as they come with a university license. Nevertheless, the costs of these licenses are \$ 45.

7.2 Results and Discussion

This section will need to have several elements, including:

- A brief summary, just a few paragraphs, of your key findings, related back to what you expected to see (essential);
- The conclusions which you have drawn from your project (essential);
- Why your project is important for researchers and practitioners (essential);
- Recommendations for future works (strongly recommended, verging on essential);
- Recommendations for practitioners (strongly recommended in management and business courses and some other areas, so check with your supervisor whether this will be expected); and a final paragraph rounding off your report or thesis.

CHAPTER 8

SUMMARY

8.1 Summary

This section will need to have :

- A brief summary, just a few paragraphs, of your key findings, related back to what you expected to see (essential);

REFERENCES

- Aggarwal, A. (2013), ‘This is how you cite a website in latex’, <http://precog.iiitd.edu.in/people/anupama>.
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- Thiruganam, M., Anuncia, S. M. and Kantipudi, S. (2010), ‘Automatic defect detection and counting in radiographic weldment images’, *International Journal of Computer Applications* **10**(2), 1–5.

LIST OF PUBLICATIONS

1. Das, Rashmi Ranjan, et al. "Adaptive predator–prey optimization for tuning of infinite horizon LQR applied to vehicle suspension system." *Applied Soft Computing* 72 (2018): 518-526.
2. Sona, D. R., Das, R. R., Dubey, S., and Kaswan, Y. (2016). PSO algorithm in quadcopter cluster for unified trajectory planning using wireless ad hoc network. In *Proceedings of the International Conference on Soft Computing Systems: ICSCS 2015, Volume 1* (pp. 819-826). Springer India.

Curriculum Vitae

Name:
Father's Name:
Date of Birth:
Nationality:
Sex:
Company Placed:
Permanent Address:
Phone number:
Mobile:
E-mail ID:

CGPA: 9.9

Examinations Taken:

1. GRE
2. TOEFL
3. GATE :99.92 Percentile

Placement Details: Placed through CDC.

Capstone Project

Project title:	Model Predictive Controller
Team Members:	
Faculty Guide:	
Semester/Year:	
Project Abstract: (Not more than 200 words)	
Project Title:	
List codes and standards that significantly affect your project (Must)	
List at least two significant realistic design constraints that are applied to your project. (Must)	
Briefly explain two significant trade-offs considered in your design, including options considered and the solution chosen (Must)	
Describe the computing aspects, if any, of your project. Specifically identifying hardware-software trade-offs, interfaces, and/or interactions	

Appendices

Appendix A

CHI SQUARE DISTRIBUTION TABLE

df	0.995	0.99	0.975	0.95	0.9	0.1	0.05	0.025	0.01	0.005
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.01	0.02	0.051	0.103	0.211	4.605	5.991	7.378	9.21	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.86
5	0.412	0.554	0.831	1.145	1.61	9.236	11.07	12.833	15.086	16.75
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.69	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.18	2.733	3.49	13.362	15.507	17.535	20.09	21.955
9	1.735	2.088	2.7	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.94	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.92	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.3
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.66	5.629	6.571	7.79	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.39	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.26	9.591	10.851	12.443	28.412	31.41	34.17	37.566	39.997
21	8.034	8.897	10.283	11.591	13.24	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	9.26	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.98	45.559
25	10.52	11.524	13.12	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.16	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.29
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.42	76.154	79.49
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.54	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

Appendix B

GANTT CHART

